

# The Fertility, Marriage, and Gender Equality Quandary

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# Motivation

- Three trends underpin the grand gender convergence in the last century:
  1. [Falling fertility](#) (Guinnane 2011)
  2. [Declining marriage](#) (Stevenson and Wolfers 2007)
  3. [Converging gender \(income\) gaps](#) (Goldin 2014)
- Many policies targeting individual outcomes
- Existing research often treats them in isolation or pairwise, lacking a unified framework

# This Paper

1. Document a **three-way trade-off**: simultaneously achieving high fertility, low single parenthood, and gender income equality is statistically unlikely
2. Develop a **unified model** of marriage, fertility, and female labor supply to explain the empirical facts and study policy implications
3. Calibrate to Mexico (1990–2015), conduct decomposition and prediction

# Key Findings

1. Mutual information analysis reveals **synergy**: the three outcomes should be analyzed jointly
2. Traditional policies cannot resolve the trade-off
3. **Redistributing childcare** to husbands can under two conditions
4. Quantitative results on Mexico from 1990 to 2015:
  - Gender-neutral TFP explains half of declining fertility and marriage
  - Gender-biased TFP and education gap reversal drive income convergence
5. Dynamic propagation through gendered impacts of single parenthood

# Literature

- Pairwise studies of fertility, marriage, and gender gaps (Galor & Weil 1996, Regalia & Ríos-Rull 2001, Greenwood et al. 2016)  
→ Establish three-way trade-off; unified framework
- Gender-biased technical change (Greenwood et al. 2005, Ngai & Petrongolo 2017)  
→ Show gender-neutral TFP also drives convergence
- Boys' disadvantage in single-parent families (Autor et al. 2019)  
→ Embed in macro model with dynamic propagation
- Policy evaluation in family economics (Doepke & Kindermann 2019)  
→ Identify policy frontier; childcare redistribution as remedy

# Roadmap

- Empirical findings
- Model
- Policy analysis
- Quantitative analysis
- Dynamic extension
- Conclusion

# Empirical Approach

1. Create datasets with fertility, dual parenthood, and gender income gap
2. Create dummy variables at sample medians:
  - High fertility (F)
  - Dual parenthood (M)
  - Gender income equality (G)
3. Visualize intersections using Venn diagrams
4. Compute mutual information to characterize dependency structure

Dartboard approach: random benchmark is  $0.5 \times 0.5 \times 0.5 = 12.5\%$

# OECD Sample

37 OECD countries, 1970–2014

721 observations

- Total fertility rate (1.69)
- Out-of-wedlock birth share (31.4%)
- Gender median earnings gap (17.2%)

Advantage: longer horizon



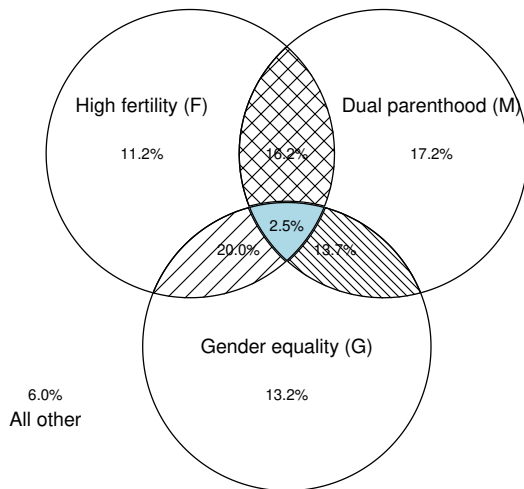
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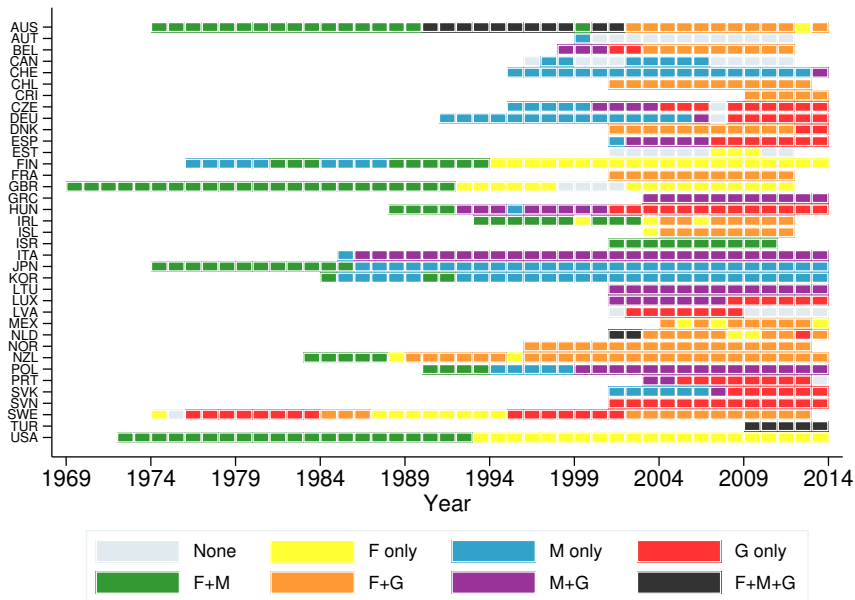
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Trinity share: 2.5%

MI analysis: synergy

# OECD Sample – Transition Paths



# World Sample

95 countries, 1990–2015

1,130 observations

- Total fertility rate (2.81)
- Children with both parents (75.9%)
- Female labor income share (32.3%)

Advantage: better measurement, wider coverage

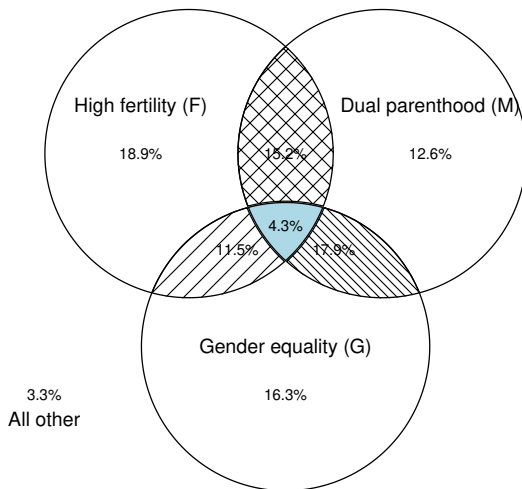
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Advantage: better measurement, wider coverage



Trinity share: 4.3%

MI analysis: synergy

# U.S. Sample (States)

50 states + D.C., 2023

- Births per 1000 women  
(56.9)
- Single-mother household  
share (17%)
- Gender earnings gap  
(25.6%)

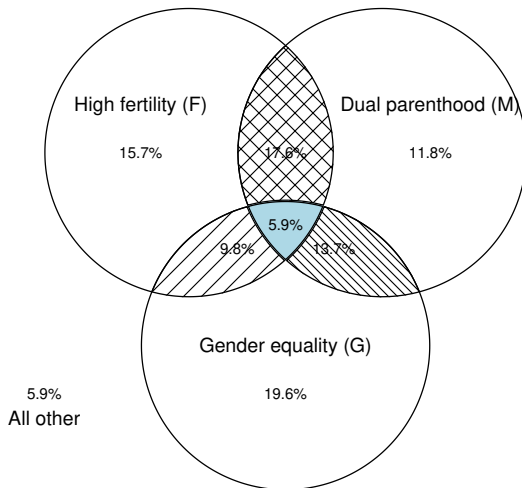
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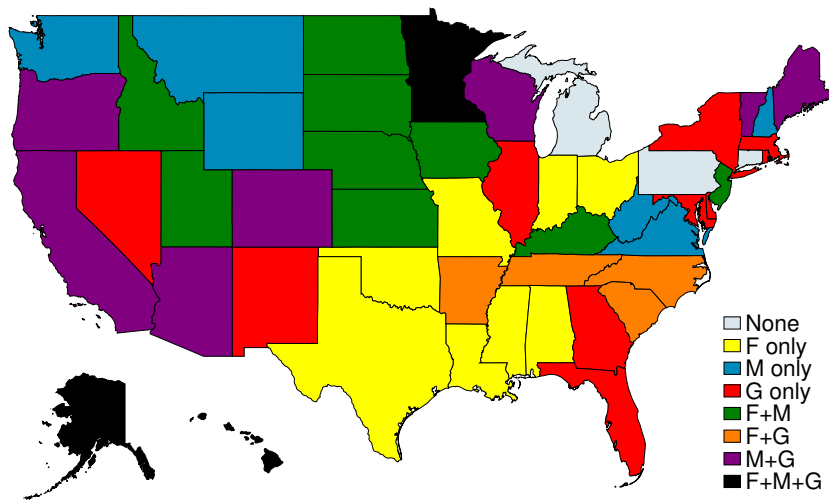
Advantage: uniform institutional setting



Trinity share: 5.9%

MI analysis: synergy

## U.S. Sample (States) – Geographic Distribution



# Questions

1. What explains the three-way trade-off?
2. Are there policies that can help escape the trade-off?
3. What role does technological progress play?



# Basic Setup

- Building on Becker (1973), Choo and Siow (2006)
- Individuals with gender  $g \in \{\sigma, \varphi\}$  and preference

$$u(c, n) = \left( (1 - \beta) \cdot c^{\frac{\rho-1}{\rho}} + \beta \cdot n^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}} \quad (1)$$

where  $\rho > 1$  so that  $u(c, 0)$  is well-defined

- Homogeneous wage within gender:  $w^\sigma$  and  $w^\varphi$
- Gender wage gap:  $\Gamma^w = w^\sigma / w^\varphi$

# Marriage and Fertility – Men

- If single, men consume labor income but have no children

$$V^{\sigma^{\uparrow},s} = u(w^{\sigma^{\uparrow}}, 0) \quad (2)$$

- Once married, husbands transfer  $\alpha$  share of income to wives

$$V^{\sigma^{\uparrow},m} = u((1 - \alpha)w^{\sigma^{\uparrow}}, n^m) \cdot \lambda \quad (3)$$

→  $\alpha$  is endogenous in marriage market equilibrium

→  $\lambda$  is the psychic benefit of marriage

- In baseline model, men bear no childcare cost

# Marriage and Fertility – Women

- Single women solve

$$V^{\text{♀},s} = \max_{c,l,n} u(c, n) \quad \text{s.t.} \quad c = w^{\text{♀}}l, \quad l = 1 - \chi n \quad (4)$$

- Married women solve

$$V^{\text{♀},m} = \max_{c,l,n} u(c, n) \cdot \lambda \quad \text{s.t.} \quad c = \alpha w^{\text{♂}} + w^{\text{♀}}l, \quad l = 1 - \chi n \quad (5)$$

- Fertility subject to wife's veto (Doepke & Kindermann 2019)  $\implies$  wives determine marital fertility

# Marriage Market Equilibrium

- Idiosyncratic taste shock  $\epsilon$  for marriage, Fréchet with scale  $\theta$
- Marriage rates:

$$\mathcal{M}^{\sigma} = \frac{1}{1 + (V^{\sigma,s}/V^{\sigma,m})^{\theta}}, \quad \mathcal{M}^{\varphi} = \frac{1}{1 + (V^{\varphi,s}/V^{\varphi,m})^{\theta}} \quad (6)$$

- Equilibrium:  $\mathcal{M}^{\sigma} = \mathcal{M}^{\varphi} = \mathcal{M}$
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- $(\alpha, n^m)$  act as market-clearing “prices”
- Lemma 1: For any wage pair  $\{w^{\sigma}, w^{\varphi}\}$ , there exists a unique equilibrium  $(\alpha, n^m)$  – intersection between  $\alpha(n^m)$  (men) and  $n^m(\alpha)$  (women)
- Lemma 2: Gains from marriage:  $V^{\varphi,m}/V^{\varphi,s} = \lambda \cdot (1 + \alpha\Gamma^w)$

# Three-Way Trade-Off: Intuition

- Equilibrium conditions reveal fundamental tension:

$$\mathcal{M} = \frac{(\lambda \cdot (1 + \alpha \Gamma^w))^\theta}{1 + (\lambda \cdot (1 + \alpha \Gamma^w))^\theta} \quad (7)$$

$$\Gamma^y = \Gamma^w / l^{\varnothing} \quad (8)$$

$$l^{\varnothing} = 1 - \chi n \quad (9)$$

- Insight:** Higher  $n$  reduces  $l^{\varnothing}$ , therefore needs low  $\Gamma^w$  to achieve gender income equality, but low  $\Gamma^w$  reduces  $\mathcal{M}$

# Traditional Policy Effects

| Policy                                      | Fertility ( $n$ ) | Marriage ( $\mathcal{M}$ ) | Gender Gap ( $\Gamma^y$ ) |
|---------------------------------------------|-------------------|----------------------------|---------------------------|
| Pro-marriage ( $\uparrow \lambda$ )         | +                 | +                          | + (widens)                |
| Gender equality ( $\uparrow w^{\text{♀}}$ ) | −                 | −                          | − (narrows)               |
| Reduce child costs ( $\downarrow \chi$ )    | +                 | +                          | + (widens)                |

- None of these policies can simultaneously improve all three outcomes
- Women's disproportionate childcare burden ties fertility to gender income gaps through labor supply  $\implies$  What if this is inefficient?

# Redistributing Childcare to Husbands

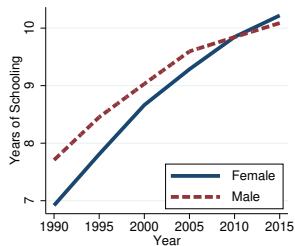
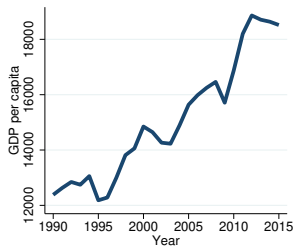
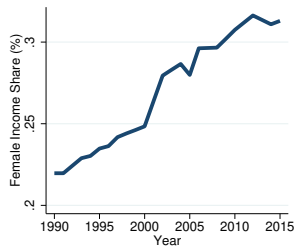
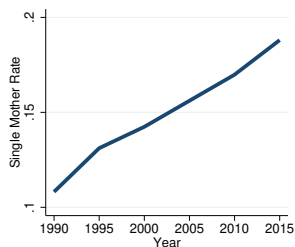
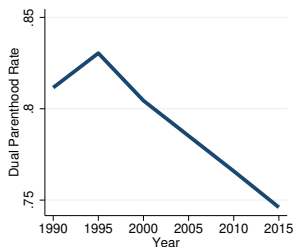
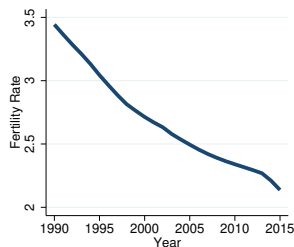
- Extension:  $\chi^{\text{♀}}$  for wives,  $\chi^{\text{♂}}$  for husbands,  $\chi = \chi^{\text{♀}} + \chi^{\text{♂}}$  for single mothers
- Wife's effective childcare cost:  $\tilde{\chi}^{\text{♀}} = \frac{\alpha w^{\text{♂}} \chi^{\text{♂}} + w^{\text{♀}} \chi^{\text{♀}}}{\alpha w^{\text{♂}} + w^{\text{♀}}}$



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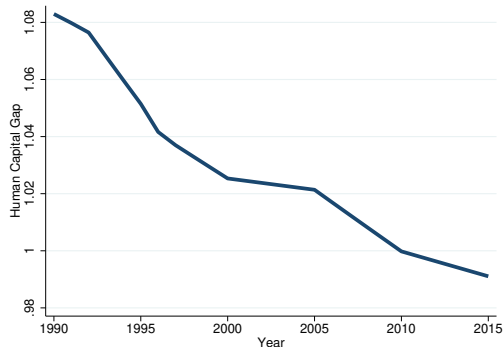
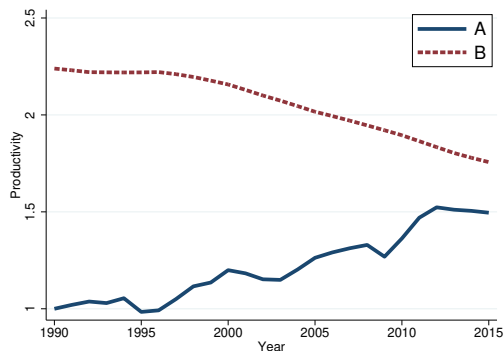
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- **Proposition:** Redistribution improves all three outcomes if:
  1. Wife's earnings exceed spousal transfers:  $w^{\text{♀}} > \alpha \cdot w^{\text{♂}}$
  2. Husband's gain from fertility exceeds consumption loss. More likely if
    - Preference weight on child  $\beta$  is large
    - Initial  $\chi^{\text{♂}}$  is small
    - Elasticity of substitution  $\rho$  is low
    - Fertility response is strong
- When both conditions hold, redistribution is **Pareto-improving**

# Mexico's Transition (1990–2015)



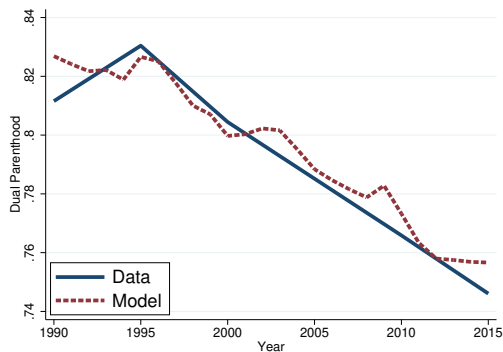
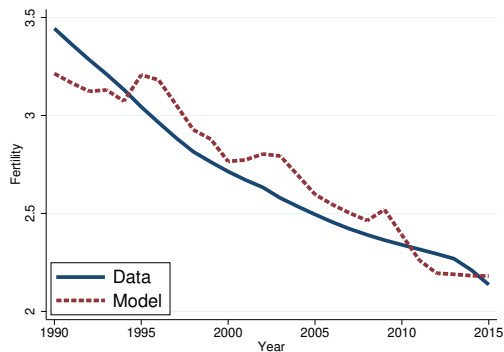
# Calibration

- Decompose wages into three components:
  - $A_t = w_t^{\sigma}$ : gender-neutral productivity
  - $\Gamma_t^h$ : gender gap in human capital
  - $B_t = (w_t^{\sigma} / w_t^{\varphi}) / \Gamma_t^h$ : gender-biased productivity



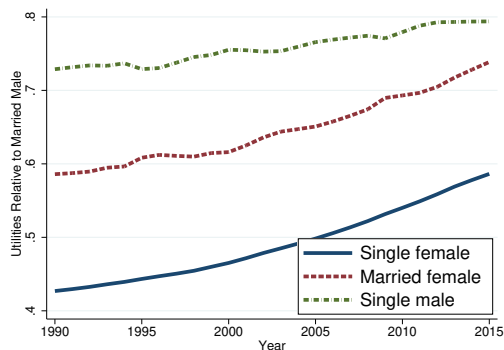
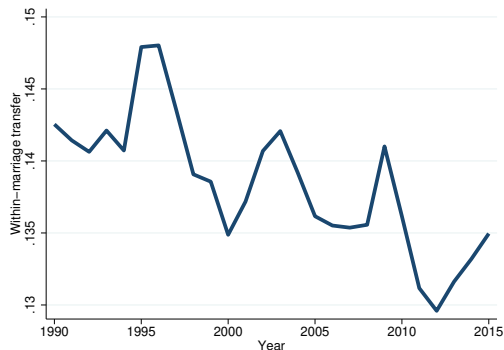
# Model Fit

- Fix  $\chi = 0.075$  (de La Croix and Doepke 2003) assuming  $\chi^{\sigma} \approx 0$
- Calibrate:  $\beta = 0.104$ ,  $\rho = 1.5$ ,  $\lambda = 1.03$ ,  $\theta = 3.4$

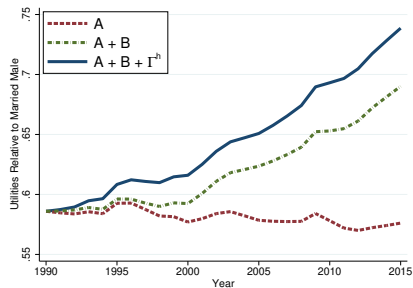
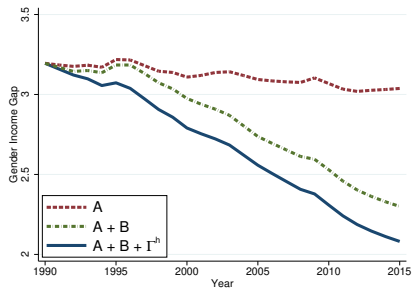
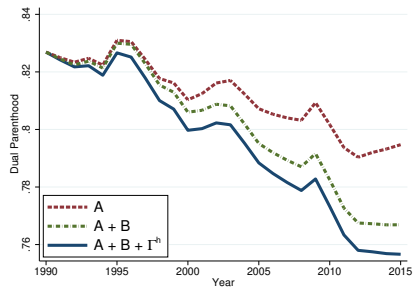
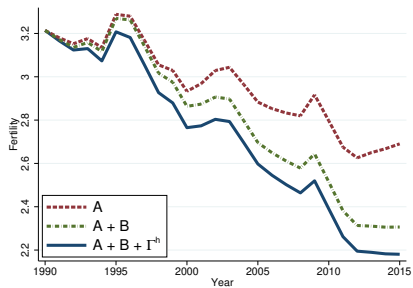


# Equilibrium Outcomes

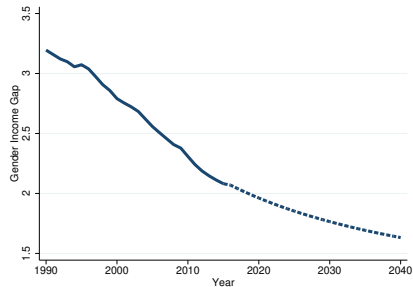
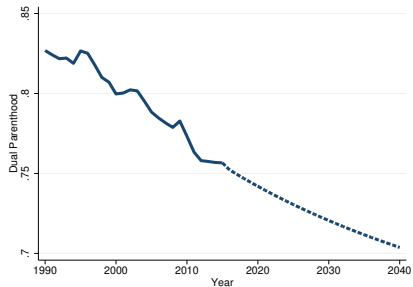
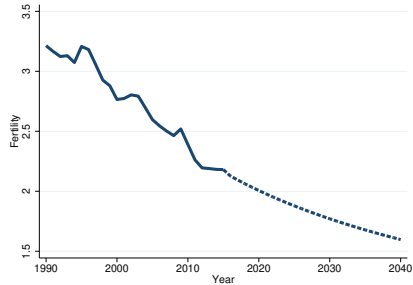
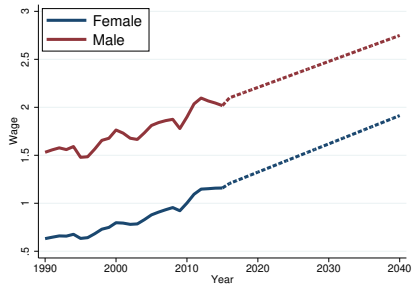
- Small reduction in  $\alpha$  despite large drop in  $n^m$  as  $w^{\text{♀}}/w^{\text{♂}}$  rises
- $\alpha \approx 0.14$  satisfies condition for wives to benefit; husband condition also satisfied  $\implies$  redistribution would be Pareto-improving



# Decomposition



# Model-Based Prediction



# Endogenous Gender Human Capital Gap

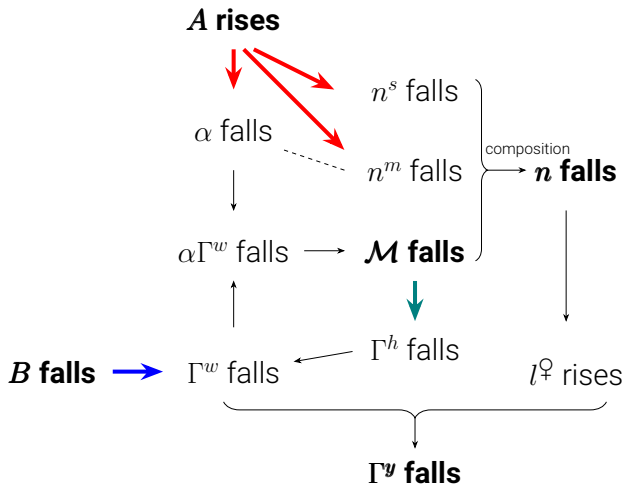
- Single parenthood affects boys and girls differently
- Evidence: Bertrand & Pan 2013, Autor et al. 2019, Wasserman 2020, Reeves 2022
- Law of motion:

$$\Gamma_{t+1}^h = \mathcal{H}(\mathcal{M}_t) \quad \text{where } \mathcal{H}'(\mathcal{M}_t) > 0 \quad (10)$$

- Creates self-reinforcing feedback: declining marriage  $\rightarrow$  lower male human capital  $\rightarrow$  further marriage decline



# Dynamic Mechanism



Red:  $A_t$  rises    Blue:  $B_t$  falls    Teal: Dynamic feedback

# Conclusion

- A three-way trade-off with synergistic dependency structure
- Unified framework explaining the empirical pattern
- Traditional policies cannot resolve the tension; redistributing childcare to husbands can under certain conditions
- Quantitative analysis: both gender-neutral and gender-biased technological change matter
- Dynamic propagation through human capital amplifies effects across generations